

Sleep Health and Lifestyle Analysis

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Agenda

Results of sleep health and lifestyle analysis

- Data Description
- Typical Amount of Physical Activity
- Number of Daily Steps
- Distribution of Heart Rates

Data Description

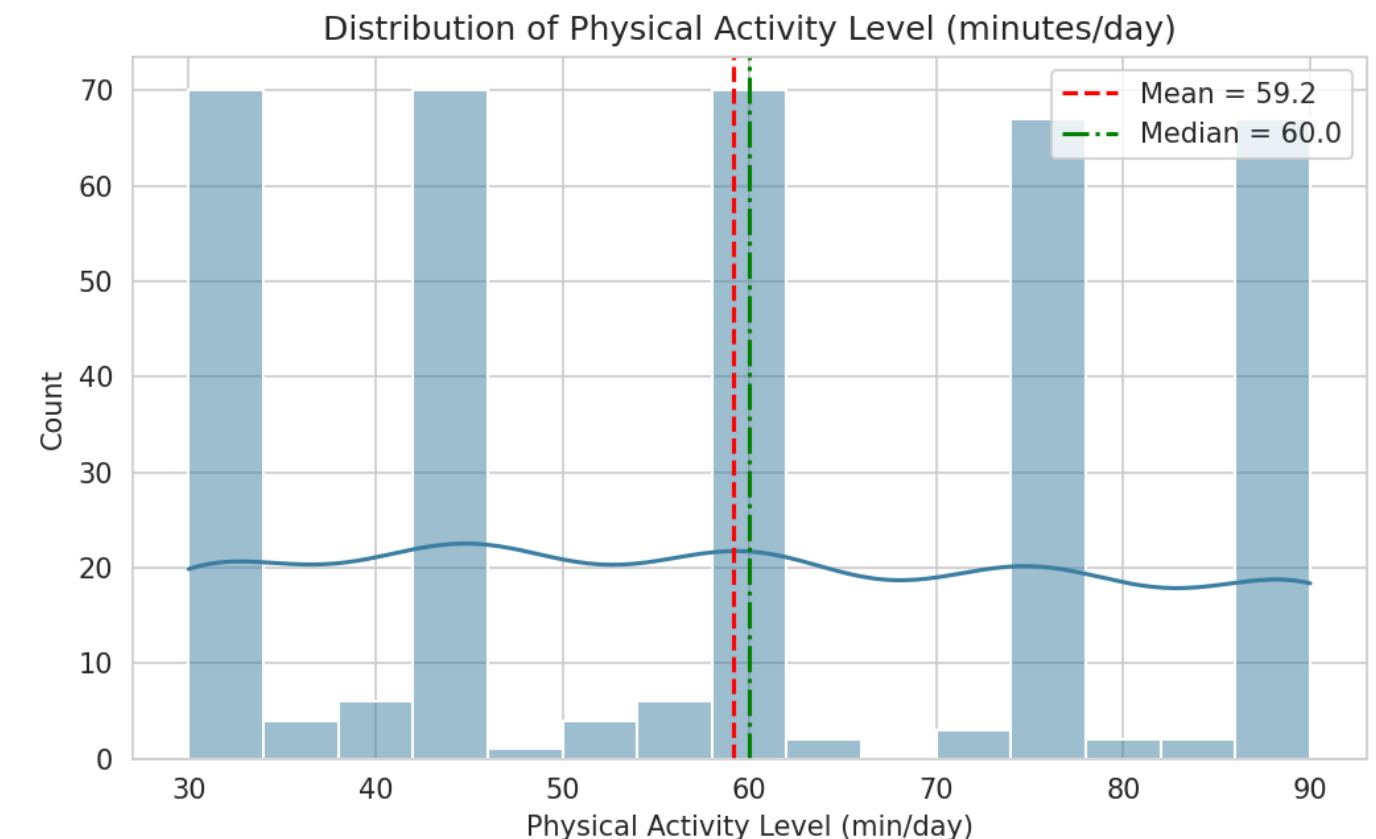
374 rows • 13 variables • Kaggle “Sleep Health and Lifestyle” dataset

- Demographics: Age (27–59), Gender, Occupation (11 roles, unevenly sampled)
- Lifestyle: Sleep Duration, Physical Activity Level, Daily Steps, Stress Level
- Health: BMI Category, Blood Pressure, Heart Rate, Sleep Disorder
- Data quality: Sleep Disorder blank for 58.6% of rows – recoded as “None” rather than dropped; “Normal” and “Normal Weight” BMI labels merged
- 64.7% of rows are duplicate lifestyle profiles under a new Person ID – effective distinct sample is closer to ~132 than 374

Typical Amount (Minutes) of Physical Activity

Median 60 min/day • roughly symmetric (skew = 0.07)

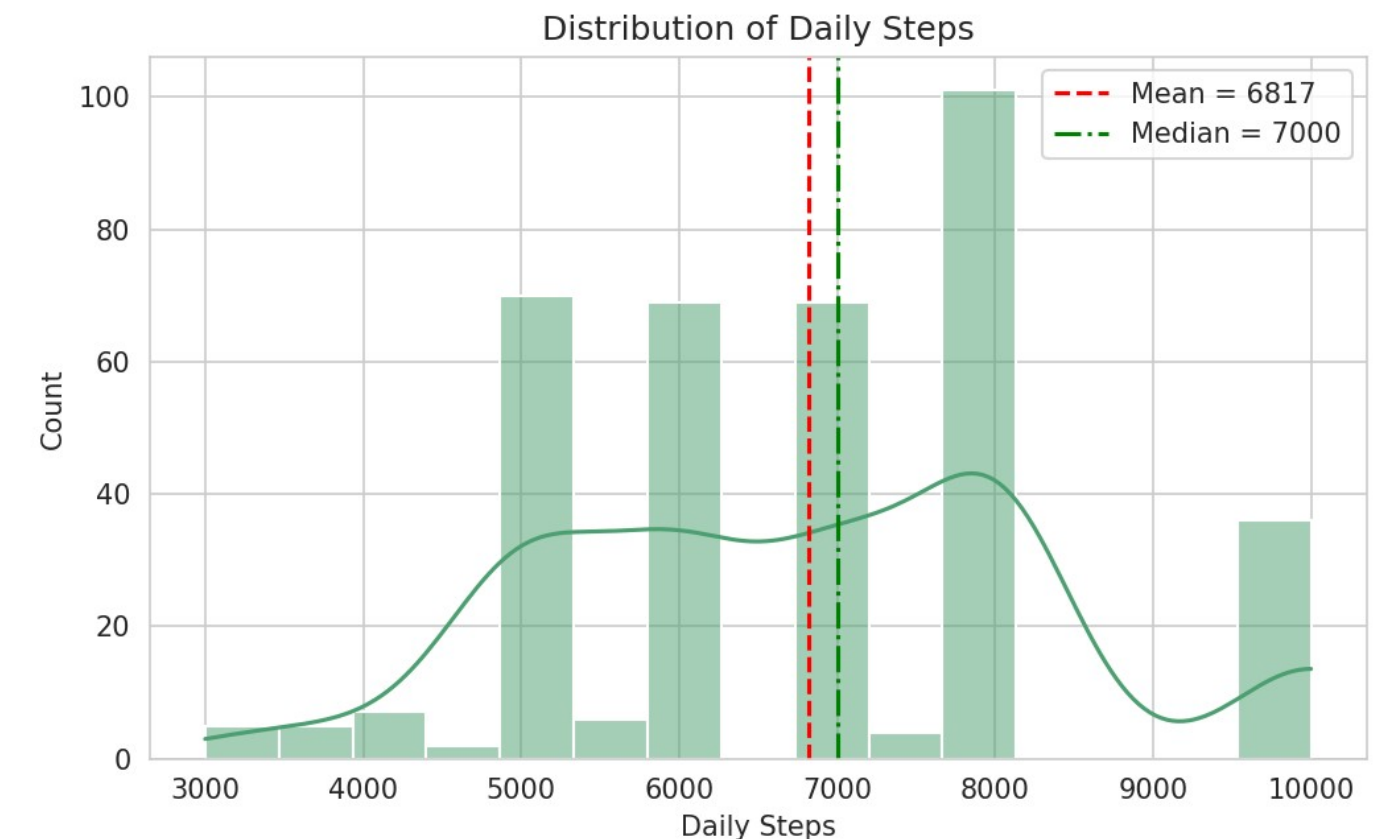
- Distribution is bimodal: a large cluster at 60–70 min/day and a smaller cluster at 30–45 min/day
- Overweight group shows the lowest median activity (45 min) vs. 60 min for both Normal and Obese groups
- Caveat: Obese subgroup is small (n=10) – treat the BMI comparison as directional, not conclusive



Analysis of Daily Steps Taken

Median ~7,000 steps/day • mild right skew (0.18)

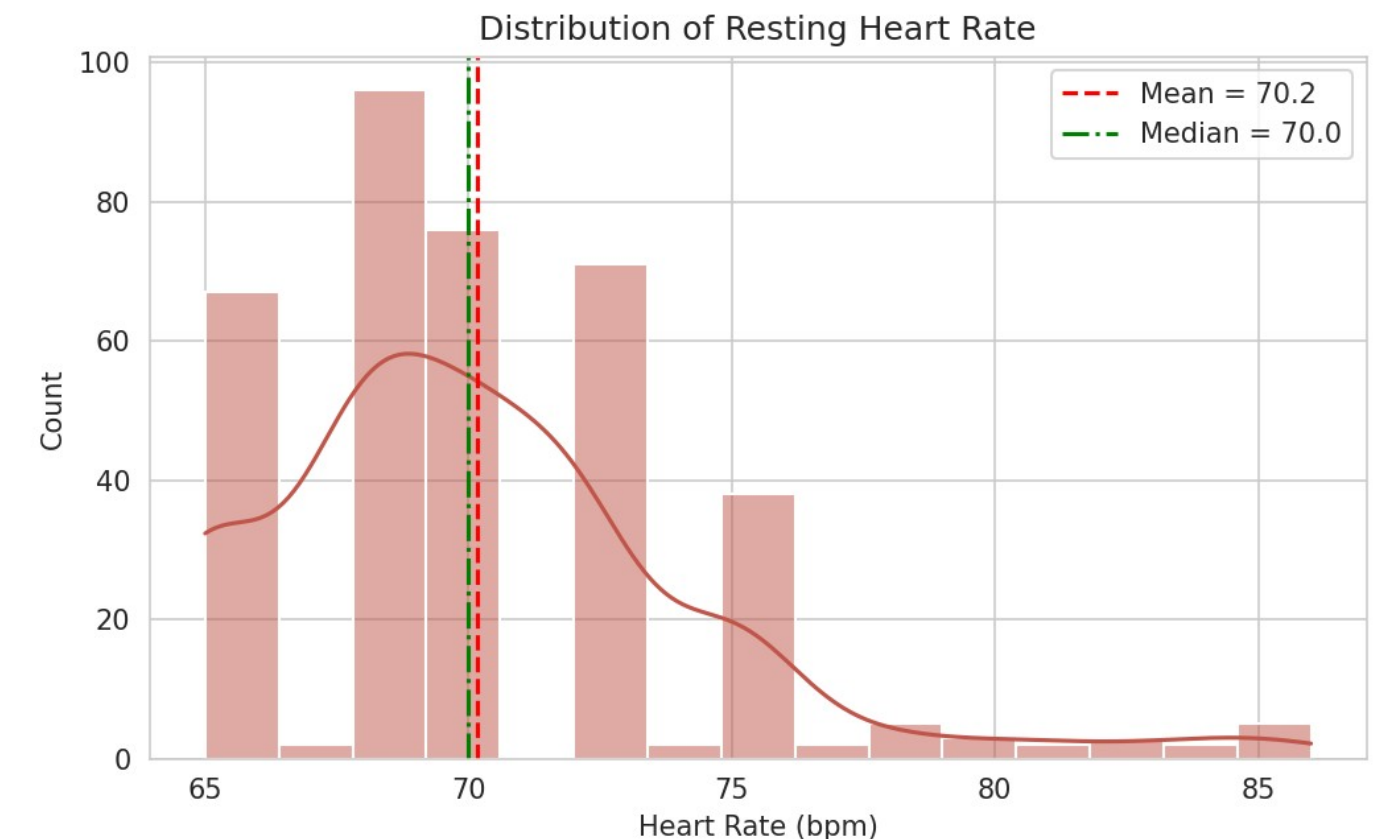
- Mean 6,817 steps/day, std dev 1,618
- Wide spread from 3,000 to 10,000 steps
- Step counts vary noticeably across occupations
- Caveat: several occupations have very small samples (e.g. n=4 Software Engineers, n=1 Manager) – not a reliable ranking of which job is “healthiest”



Distribution of Heart Rates

Median 70 bpm • most skewed variable in the dataset (skew = 1.23)

- Right-skewed distribution: most respondents cluster in the low-to-mid 60s–70s bpm with a long tail of higher readings
- Because it's skewed, median (not mean) is the correct “typical value” to report
- Sleep Apnea (median 75 bpm) and Insomnia (median 72 bpm) groups run higher than the no-disorder group (median 70 bpm) – consistent with known links between sleep disorders and elevated resting heart rate



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